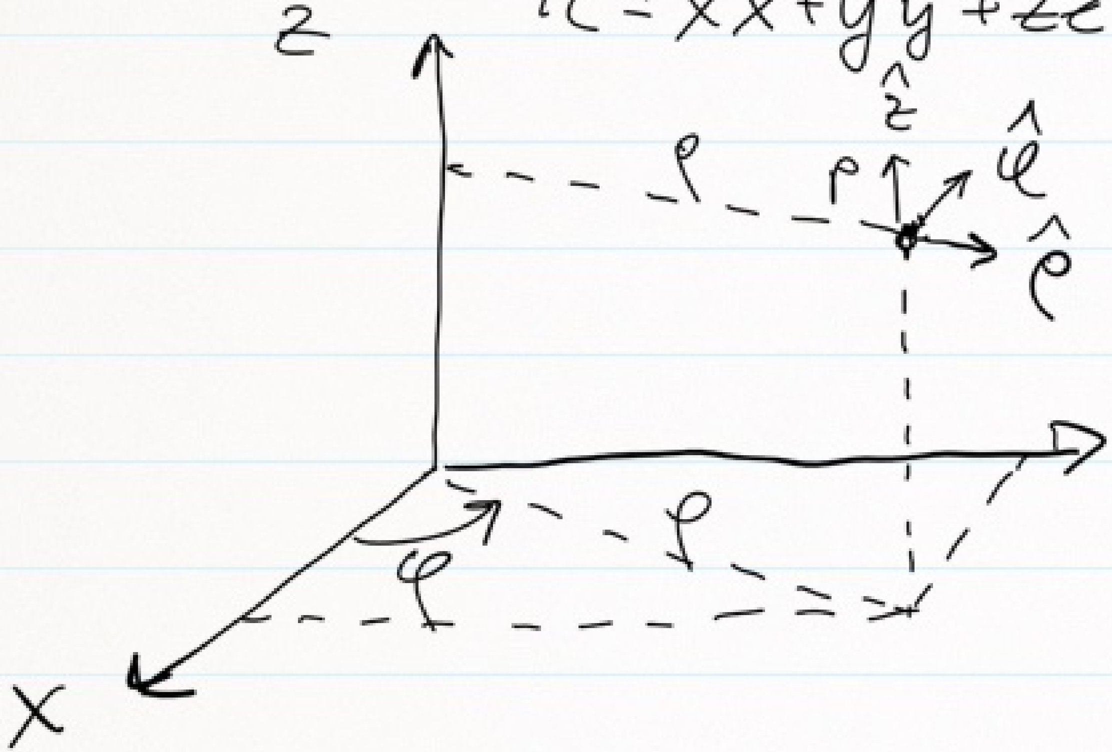
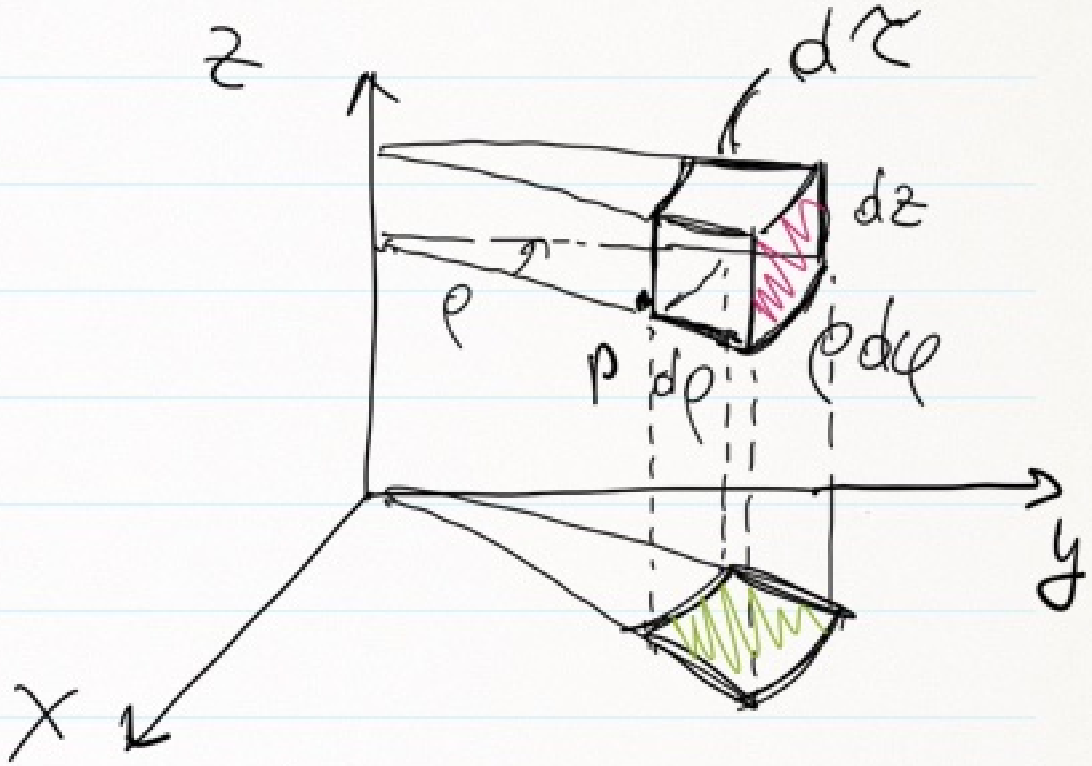
23/09/20

COORDINATE CILINDRICHE

$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$





- ρ : $0 \leq \rho < \infty$
- φ : $0 \leq \varphi \leq 2\pi$
- z : $-\infty < z < +\infty$

$$\begin{cases} x = \rho\cos\varphi \\ y = \rho\sin\varphi \\ z = z \end{cases}$$

$$\hat{\rho} = \frac{\partial\vec{r}}{\partial\rho} \Big/ \left|\frac{\partial\vec{r}}{\partial\rho}\right|$$

$$\hat{\varphi} = \frac{\partial\vec{r}}{\partial\varphi} \Big/ \left|\frac{\partial\vec{r}}{\partial\varphi}\right|$$

$$\hat{z} = \frac{\partial\vec{r}}{\partial z} \Big/ \left|\frac{\partial\vec{r}}{\partial z}\right|$$

$$d\vec{l} = d\rho\,\hat{\rho} + \rho\, d\varphi\,\hat{\varphi} + dz\,\hat{z}, \quad d\tau = \rho\, d\rho\, d\varphi\, dz$$

$$\rho = \sqrt{x^2 + y^2} \ , \quad \varphi = \operatorname{atan}\left(\frac{y}{x}\right) \ , \quad z = z$$

$$(u, v, w) \longrightarrow (u + du, v + dv, w + dw)$$

$$d\vec{l} = f \, du \, \hat{u} + g \, dv \, \hat{v} + h \, dw \, \hat{w} \quad \leftarrow$$

	u	v	w	f	g	h
CART.	x	y	z	1	1	1
SFER.	r	θ	φ	1	r	$r \sin\theta$
CIL.	ρ	φ	z	1	ρ	1

$$t(u, v, w) \ , \quad dt = \frac{\partial t}{\partial u} du + \frac{\partial t}{\partial v} dv + \frac{\partial t}{\partial w} dw$$

$$\shortparallel$$

$$dt = \vec{\nabla} t \cdot d\vec{l} = (\vec{\nabla} t)_u \cdot f du + (\vec{\nabla} t)_v \cdot g dv + (\vec{\nabla} t)_w \cdot h dw$$

$$(\vec{\nabla} t)_u \equiv \frac{1}{f} \frac{\partial t}{\partial u} \ , \quad (\vec{\nabla} t)_v \equiv \frac{1}{g} \frac{\partial t}{\partial v} \ , \quad (\vec{\nabla} t)_w \equiv \frac{1}{h} \frac{\partial t}{\partial w}$$

OPERATORE <u>NABLA</u>

$$\vec{\nabla} \equiv \left(\frac{1}{f}\frac{\partial}{\partial u}, \frac{1}{g}\frac{\partial}{\partial v}, \frac{1}{h}\frac{\partial}{\partial w} \right)$$

$$\vec{\nabla}_{cil} \equiv \left(\frac{\partial}{\partial \rho}, \frac{1}{\rho}\frac{\partial}{\partial \varphi}, \frac{\partial}{\partial z} \right) = \frac{\partial}{\partial \rho}\hat{\rho} + \frac{1}{\rho}\frac{\partial}{\partial \varphi}\hat{\varphi} + \frac{\partial}{\partial z}\hat{z}$$

<u>DIVERGENZA</u>

$$\vec{F} = F_\rho \hat{\rho} + F_\varphi \hat{\varphi} + F_z \hat{z}$$

$$\vec{\nabla} \cdot \vec{F} = ?$$

$$\left(\frac{\partial}{\partial \rho}, \frac{1}{\rho}\frac{\partial}{\partial \varphi}, \frac{\partial}{\partial z} \right) \cdot \vec{F} =$$

prima DERIVO, poi MOLTIPLICO

$$= \hat{\rho} \cdot \frac{\partial \vec{F}}{\partial \rho} + \frac{\hat{\varphi}}{\rho} \cdot \frac{\partial \vec{F}}{\partial \varphi} + \hat{z} \cdot \frac{\partial \vec{F}}{\partial z} =$$

$= \ldots$

$$\boxed{\vec{\nabla} \cdot \vec{F} = \frac{1}{\rho} \frac{\partial}{\partial \rho}\left(F_\rho \rho\right) + \frac{1}{\rho} \frac{\partial F_\varphi}{\partial \varphi} + \frac{\partial F_z}{\partial z}}$$

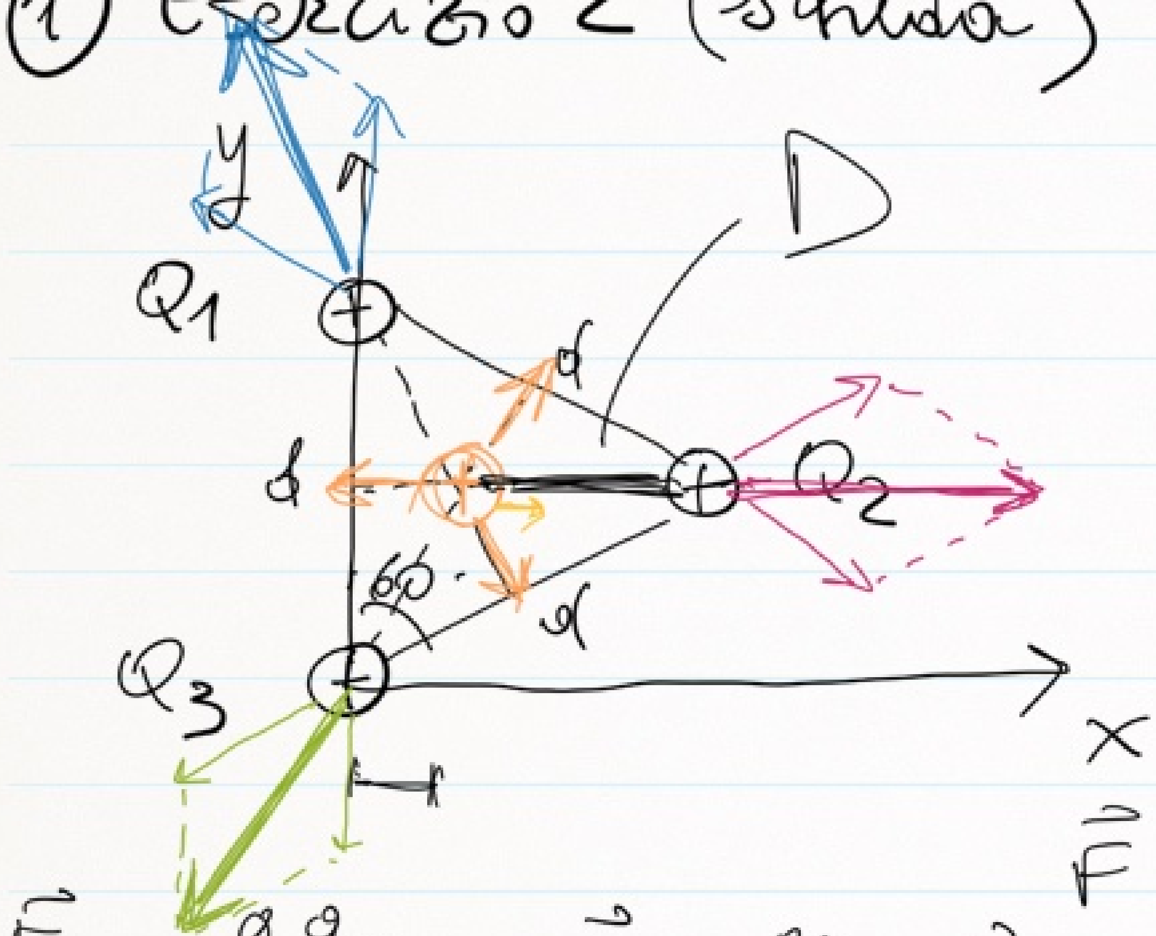
FORMULA <u>ROTORE</u> :

$$\vec{\nabla} \times \vec{F} = \hat{\rho}\left[\frac{1}{\rho} \frac{\partial F_z}{\partial \varphi} - \frac{\partial F_\varphi}{\partial z}\right] +$$

$$\hat{\varphi}\left[\frac{\partial F_\rho}{\partial z} - \frac{\partial F_z}{\partial \rho}\right] +$$

$$\hat{z}\left[\frac{1}{\rho} \frac{\partial}{\partial \rho}\left(F_\varphi \cdot \rho\right) - \frac{1}{\rho} \frac{\partial F_\rho}{\partial \varphi}\right]$$

① Esercizio 2 (scheda)

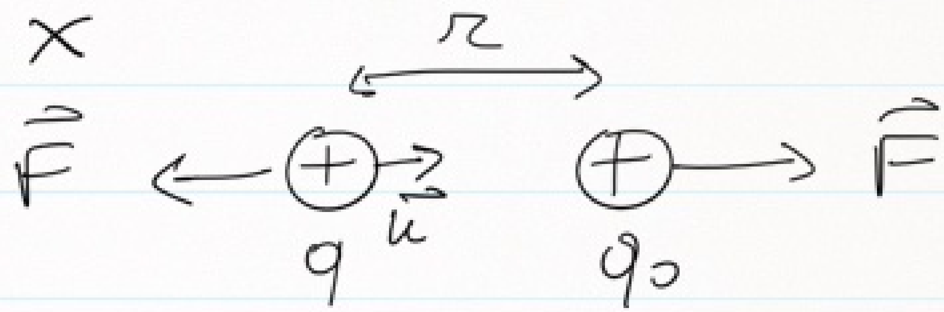


Q_1 @ $P_1(0, d)$

Q_2 @ $P_2\left(\frac{\sqrt{3}}{2}d, \frac{d}{2}\right)$

Q_3 @ $P_3(0, 0)$

Q_4 @ $P_4\left(\frac{d}{2\sqrt{3}}, \frac{d}{2}\right)$



$$\vec{F} = \frac{q q_0}{4\pi\varepsilon_0 r^2}\vec{u} = \frac{q q_0}{4\pi\varepsilon_0 r^3}\vec{r}$$

$K = \{1, 2, 3\}$ $J = \{x, y, z\}$

$$F_{KJ}(x,y,z) = \frac{Q_K}{4\pi\varepsilon_0}\sum_i{}' \frac{Q_i\,(J_K - J_i)}{\left[(x_K-x_i)^2+(y_K-y_i)^2+(z_K-z_i)^2\right]^{3/2}}$$

PRINCIPIO DI SOVRAPPOSIZIONE EFFETTI

$$F_{2x} = \frac{Q_2}{4\pi\varepsilon_0}\left[\frac{Q_1(x_2-x_1)}{d^3} + \frac{Q_3(x_2-x_3)}{d^3}\right] =$$

$$= \frac{Q_2}{4\pi\varepsilon_0}\left[\frac{Q_1}{d^3}\left(\frac{\sqrt{3}}{2}d - 0\right) + \frac{Q_3}{d^3}\left(\frac{\sqrt{3}}{2}d - 0\right)\right]$$

$$= \frac{Q^2}{4\pi\varepsilon_0} \cdot \frac{\sqrt{3}}{d^2}$$

$$F_{2y} = \frac{Q_2}{4\pi\varepsilon_0}\left[\frac{Q_1(y_2-y_1)}{d^3} + \frac{Q_3(y_2-y_3)}{d^3}\right] = 0$$

$$\vec{F}_2 = F_{2x}\hat{x} + \cancel{F_{2y}}\,\hat{y} = \frac{Q^2}{4\pi\varepsilon_0 d^2}\sqrt{3}\,\hat{x}$$

$$F_{3x} = \frac{Q_3}{4\pi\varepsilon_0}\left[\cancel{\frac{Q_1(x_3-x_1)}{d^3}} + \frac{Q_2(x_3-x_2)}{d^3}\right] =$$

$$= -\frac{Q^2}{4\pi\varepsilon_0}\cdot\frac{\sqrt{3}}{2d^2}$$

$$F_{3y} = \frac{Q_3}{4\pi\varepsilon_0}\left[\frac{Q_1(y_3-y_1)}{d^3} + \frac{Q_2(y_3-y_2)}{d^3}\right]$$

$$= -\frac{Q^2}{4\pi\varepsilon_0}\cdot\frac{3}{2d^2}$$

$$\vec{F}_3 = F_{3x}\,\hat{x} + F_{3y}\,\hat{y} = \frac{Q^2}{4\pi\varepsilon_0 d^2}\left[-\frac{\sqrt{3}}{2}\hat{x} - \frac{3}{2}\hat{y}\right]$$

$$F_{1x} = -\frac{Q^2}{4\pi\varepsilon_0}\frac{\sqrt{3}}{2d^2}$$

$$F_{1y} = \frac{Q^2}{4\pi\varepsilon_0}\frac{3}{2d^2}$$

$$\vec{F}_1 = F_{1x}\hat{x} + F_{1y}\hat{y}$$

$$= \frac{Q^2}{4\pi\varepsilon_0 d^2}\left[-\frac{\sqrt{3}}{2}\hat{x} + \frac{3}{2}\hat{y}\right]$$

(b) $D = \frac{d}{\sqrt{3}}$

$$F_{4x} = \frac{Q_4}{4\pi\varepsilon_0}\left[\frac{Q_1\left(\frac{d}{2\sqrt{3}} - 0\right)}{D^3} + \frac{Q_2\left(\frac{d}{2\sqrt{3}} - \frac{\sqrt{3}d}{2}\right)}{D^3} + \frac{Q_3\left(\frac{d}{2\sqrt{3}} - 0\right)}{D^3}\right] = 0$$

OSSERVAZIONE: quanto vale il campo in P_4?

$$F_{4y} = 0 \quad \Rightarrow \quad \vec{F}_4 = \vec{0}$$

(c) Q_4 @ $P_4^* \left(\frac{d}{2\sqrt{3}} + \varepsilon, \frac{d}{2} \right)$

$$D \rightarrow D^* = D_\varepsilon$$

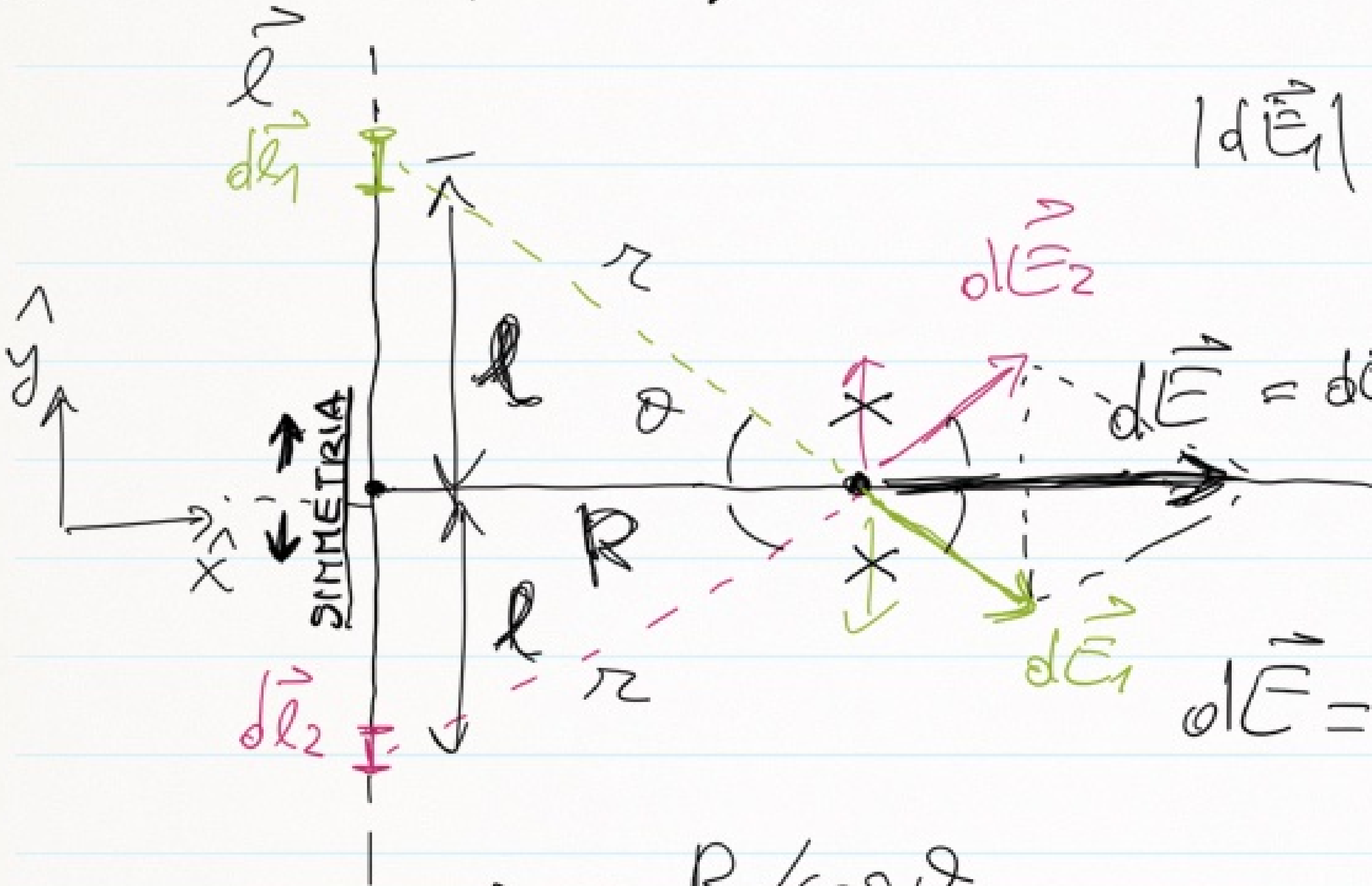
$$F_{4x}^* = -\frac{Q^2}{4\pi\varepsilon_0 D_\varepsilon^3} \cdot 3\varepsilon \rightarrow -\frac{Q^2}{4\pi\varepsilon_0 R^3} x$$

$$m \frac{d^2 x}{dt^2} = F_{4x}^* \Rightarrow m\ddot{x} + \frac{Q^2}{4\pi\varepsilon_0 R^3} x = 0$$

$$\ddot{x} + \omega_0^2 x = 0 \Rightarrow \omega_0 = \sqrt{\frac{Q^2}{4\pi\varepsilon_0 R^3 m}}$$

$$T = \frac{2\pi}{\omega_0}$$

② Esercizio 4 (scheda) $[\lambda] = C/m$, $dQ = \lambda\, dl$



$$|d\vec{E}_1| = |d\vec{E}_2| = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dl}{r^2}$$

$$d\vec{E} = d\vec{E}_1 + d\vec{E}_2$$

SOVRAPPOSIZIONE DEGLI EFFETTI

$$d\vec{E} = 2 \cdot \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dl}{r^2}\cos\theta\, \hat{x}$$

$$\to r = R/\cos\theta$$

(a)

$$l = R \cdot \tan\theta \to dl = R\, d\theta / \cos^2\theta$$

$$\vec{E} = \int d\vec{E} = \frac{\hat{x}}{2\pi\varepsilon_0}\int_0^{\pi/2}\frac{\lambda}{R}\cos\theta\, d\theta = \frac{\hat{x}}{2\pi\varepsilon_0}\frac{\lambda}{R}[\sin\theta]_0^{\pi/2} =$$

$$= \frac{\hat{x}}{2\pi\varepsilon_0}\cdot\frac{\lambda}{R}$$

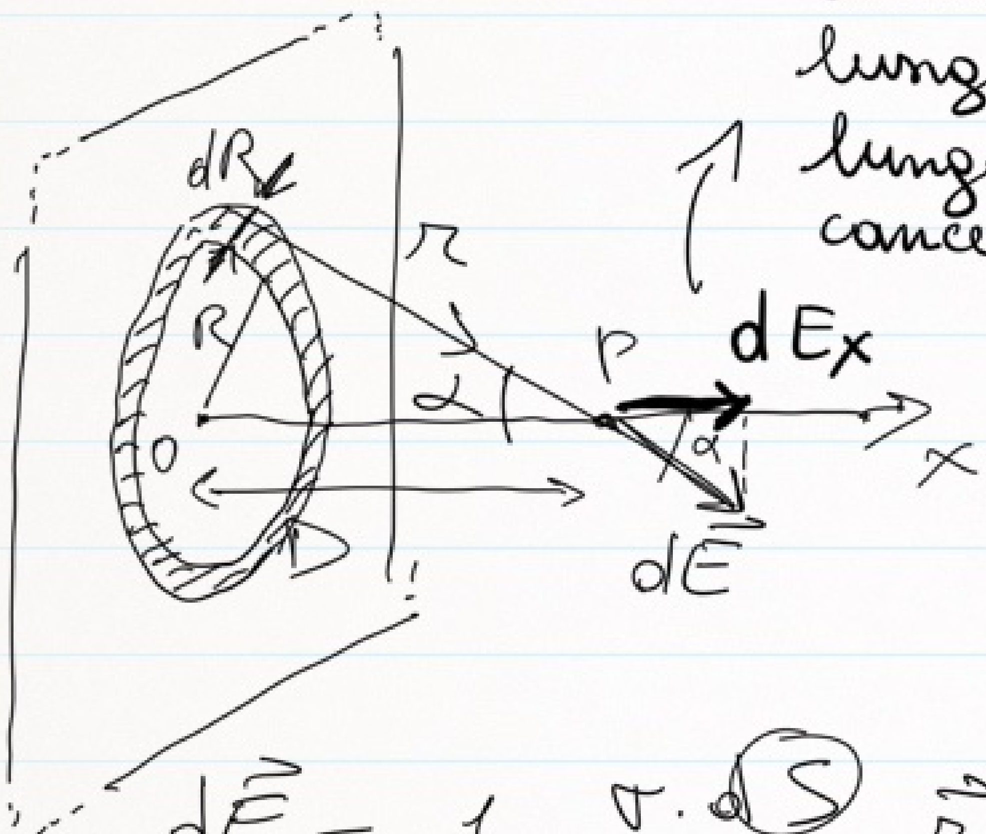
(b) @ $y = l$ $\quad \theta_1 = \arcsin\left(\frac{l}{r}\right) = \arcsin\left(\frac{l}{\sqrt{R^2+l^2}}\right)$

$$\vec{E} = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{R}\left[\sin\theta\right]_0^{\theta_1} = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{R}\frac{\ell}{\sqrt{R^2+\ell^2}}$$

(3) Esercizio 7 & 8 (scheda)

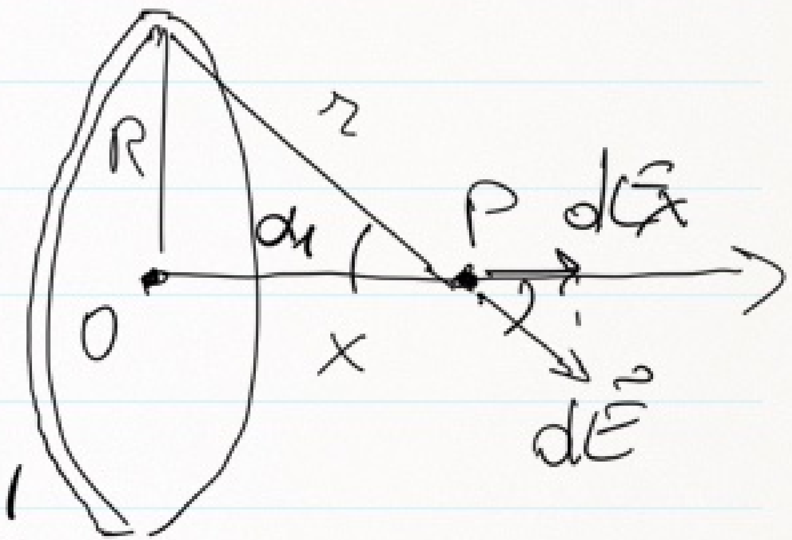
Es. 8

N.B: "sopravvivono" solo i contributi lungo x, quelli lungo y si cancellano!



Fare sempre attenzione alle simmetrie!

ES. 7



$$d\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{\sigma \cdot dS}{r^3}\vec{r}$$

$dS = 2\pi R\, dR$

$r = D/\cos\alpha$

$R = D \tan\alpha$

$dR = D \dfrac{d\alpha}{\cos^2\alpha}$

sostituire questi

$dE_x = dE \cdot \cos\alpha$

$\displaystyle \stackrel{!}{=} \frac{\sigma}{2\varepsilon_0} \sin\alpha\, d\alpha$

$$E = \int_0^{\pi/2} \frac{\sigma}{2\varepsilon_0} \sin\alpha\, d\alpha =$$

$$= \frac{\sigma}{2\varepsilon_0} \left[-\cos\alpha\right]_0^{\pi/2} = \sigma/2\varepsilon_0$$

$$\boxed{\vec{E} = \hat{x}\, \sigma/2\varepsilon_0}$$

CAMPO UNIFORME!

Nel caso del disco sottile, l'impostazione della soluzione è quindi uguale fino a quando arriva il momento di integrare

OSSERVAZIONI SUL DISCO

1) campo per $x \to 0$?

2) " per $x \gg R$?

$$\alpha_1 = \arccos\left(\frac{x}{\sqrt{x^2+R^2}}\right)$$

$$\boxed{\vec{E} = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]\hat{x}}$$